UNIT 3

Exercise 9 : AI assistant answering questions using external documents

Aim

To generate numerical embeddings for text and identify semantically similar sentences using embedding similarity.

Algorithm

- Start the program.
- Load a pretrained sentence-embedding model.
- Define a collection of text documents.
- Convert the documents into numerical vectors.
- Convert the user's query into an embedding.
- Calculate similarity scores between the query and documents.
- Display the most similar document.
- Stop.

Python Code and Output

```
q_vector = embedding_model.encode(
    [question], normalize_embeddings=True
)[0]

scores = doc_vectors @ q_vector
best_indices = np.argsort(scores)[::-1][:2]

context = "\n".join(
    external_docs[i] for i in best_indices
)

answer = generate_answer(question, context)

print("\nRetrieved document(s):")
for i in best_indices:
    print("-", uploaded.keys().__iter__().__next__()
          if len(external_docs) == 1 else f"Document {i + 1}")

print("\nAI Assistant:", answer)
```

```
Choose Files ai_notes.txt
ai_notes.txt(text/plain) - 70 bytes, last modified: 10/27/2026 - 100% done
Saving ai_notes.txt to ai_notes (1).txt
Loaded: ai_notes (1).txt
Ask a question about your documents: What is artificial Intelligence?

Retrieved document(s):
- ai_notes (1).txt

AI Assistant: enables machines to perform intelligent tasks
```

Result

Text embeddings were generated successfully, and semantic similarity search identified the most relevant document.